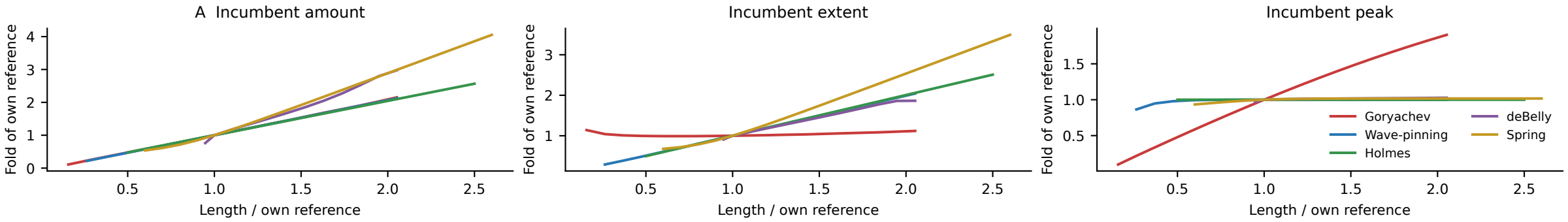
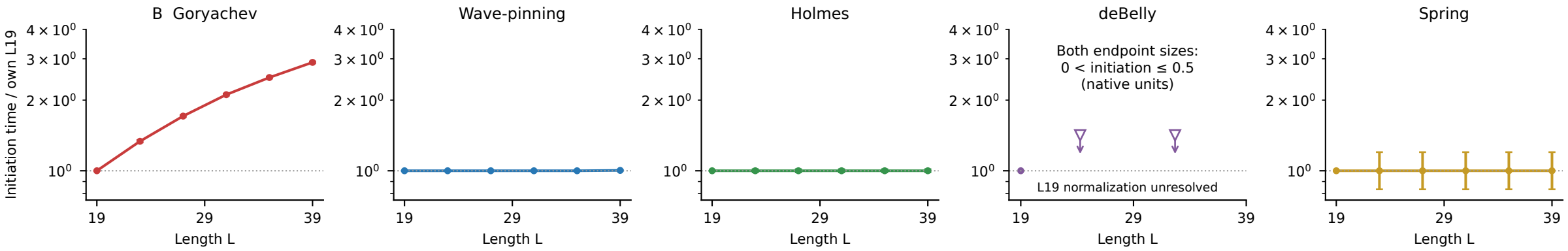
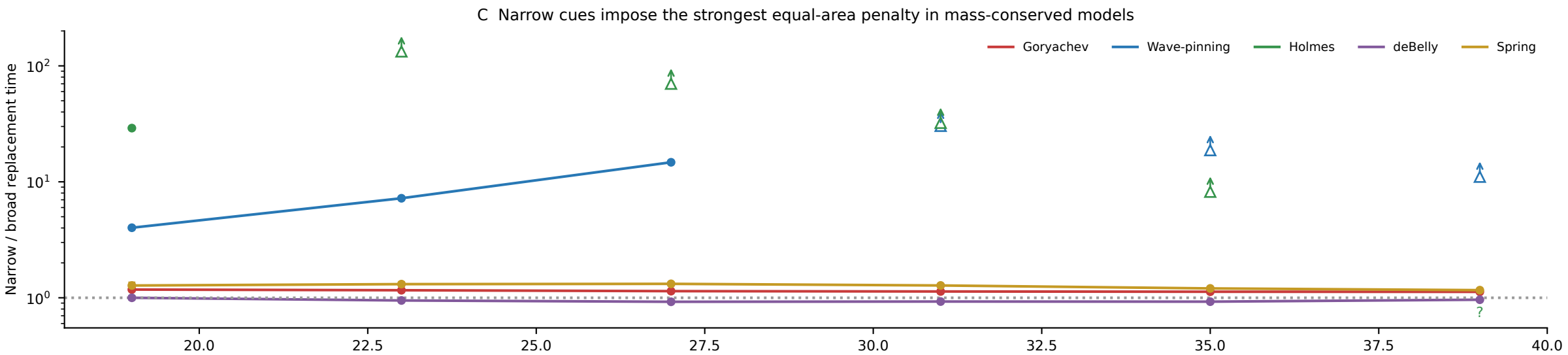
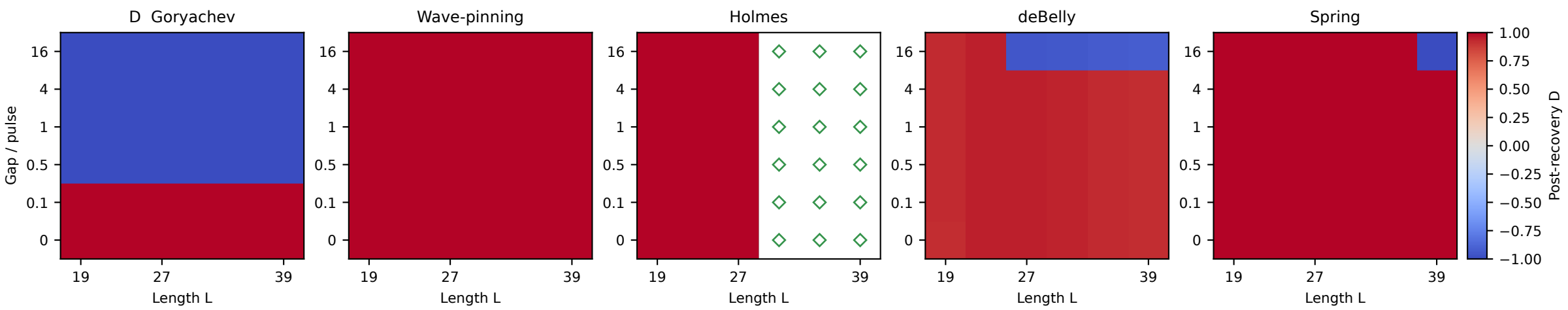
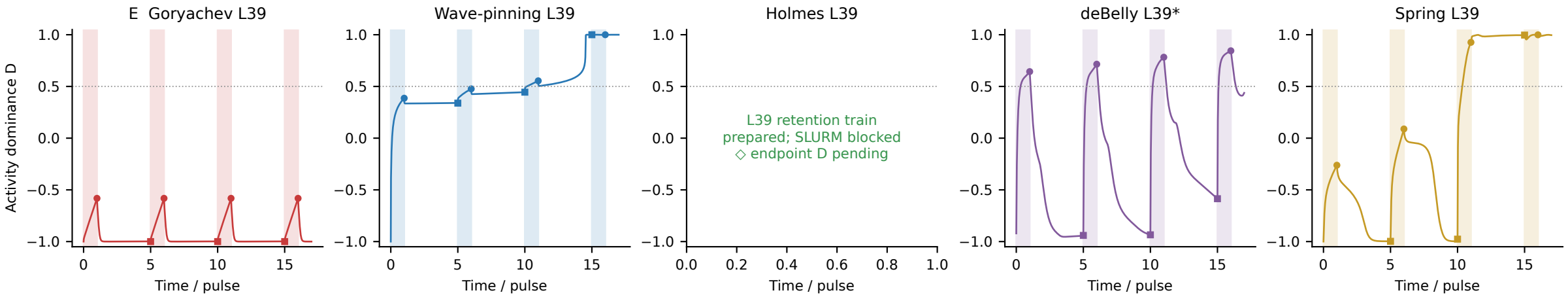
Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures











F Shape and timing signatures distinguish proposed mechanism classes

Class / models	Initiation slows	Equal-area penalty	Share through gaps	Commit after collapse	Measurement
Initiation-limited Goryachev	Yes	Modest (~1.1×)	Resets	No	Beam width at fixed dose; pulse trains; gap-end share
Mass-conserved, saturating Wave-pinning, Holmes	No	Yes; Holmes growth unresolved at L39	Ratchet; Holmes large L pending	—	Beam width at fixed dose; pulse trains; gap-end share
Slow non-activity state deBelly*, Spring	No resolved slowing*	Modest (~1×)	Collapses early	Yes	Beam width at fixed dose; pulse trains; gap-end share

◇ Holmes large-L retention is unmeasured; ? width ratio uninformative. Open triangles + arrows denote bounds.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).

Separation slows the resolution of coexisting regions more than it slows driven takeover

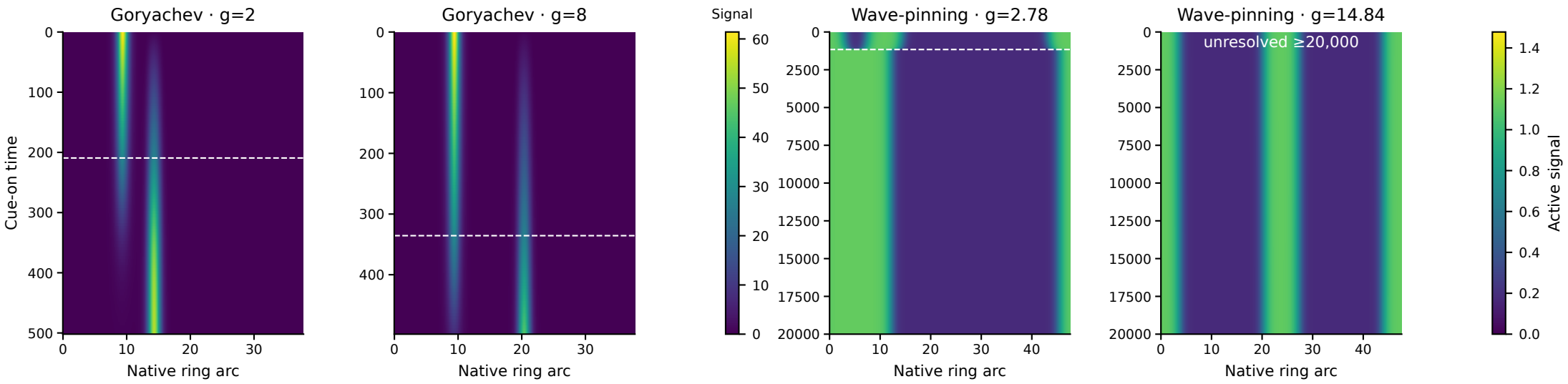
A Rotating a cue separates distance from native cell size



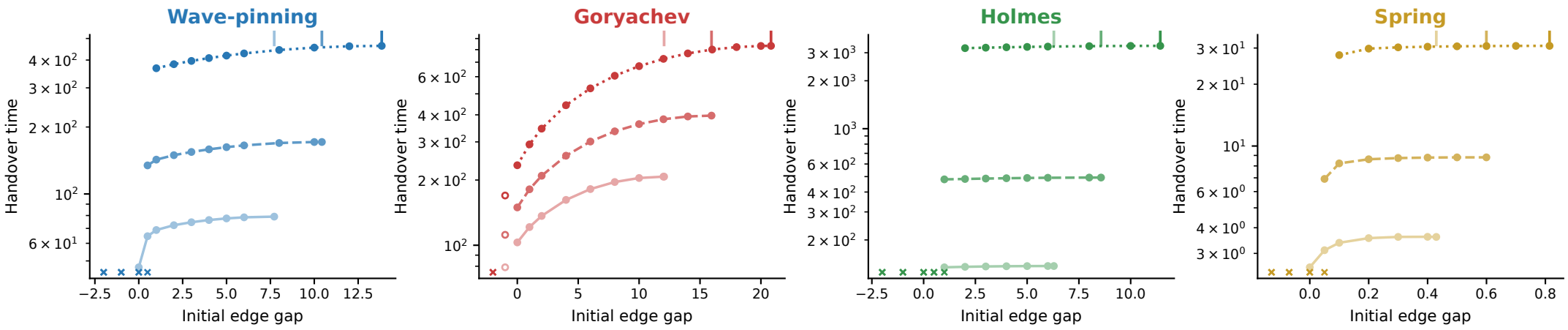
Same physical gap across sizes

A1: continuous cue → handover
B: equal complete-region amounts,
cue withdrawal → single-peak resolution
Blue incumbent; orange cue; g is edge gap.

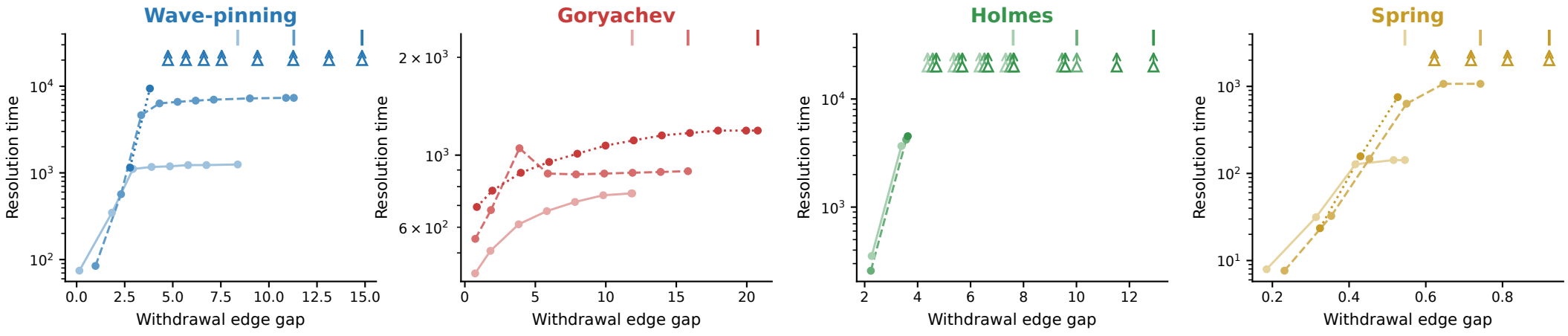
B Native histories: driven handover (left); time after withdrawal (right)



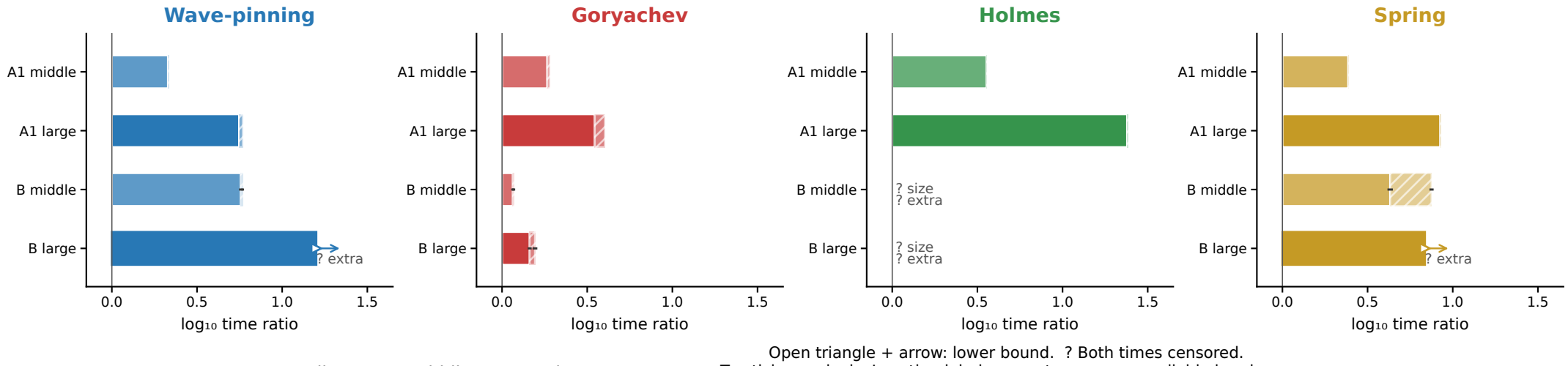
C Driven handover against initial edge gap



D Cue-free resolution against measured withdrawal gap · equal amounts



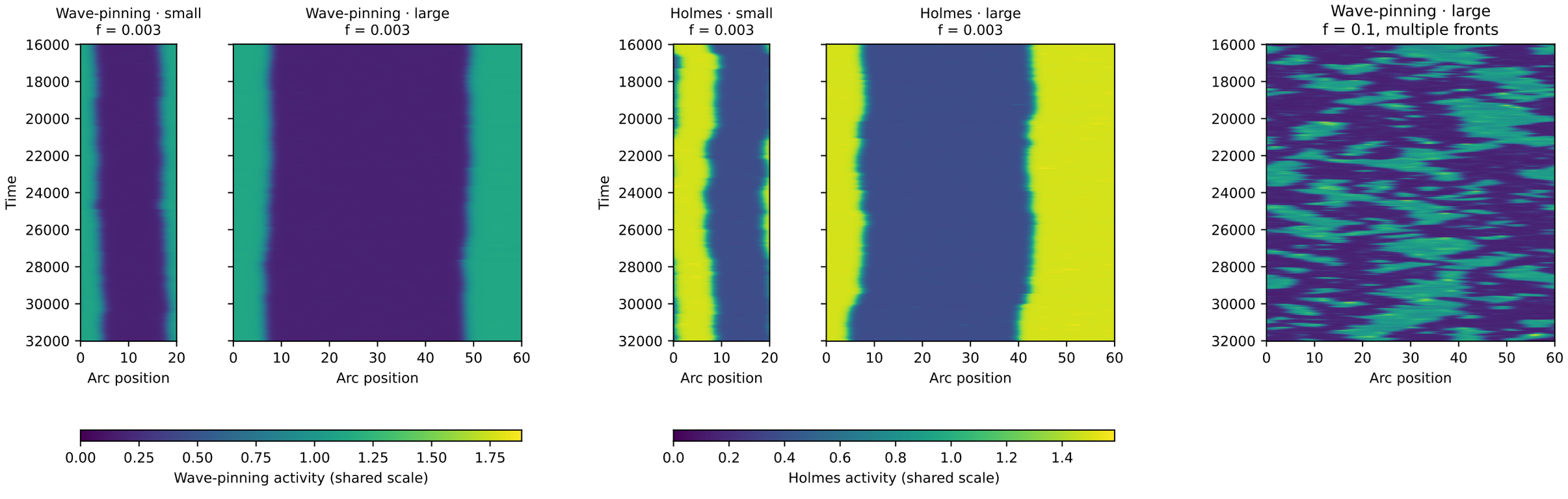
E Size at matched gap (solid) + extra separation (hatched)

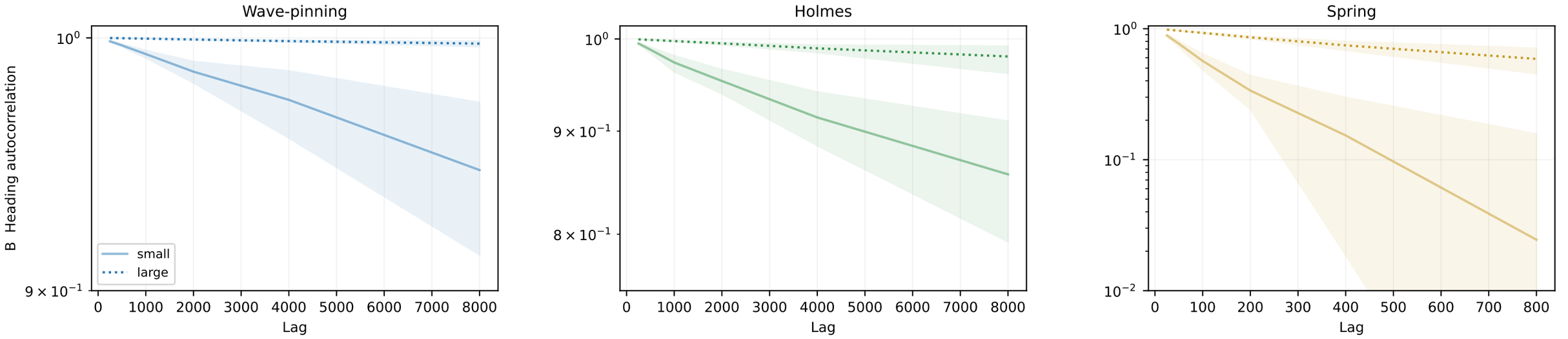


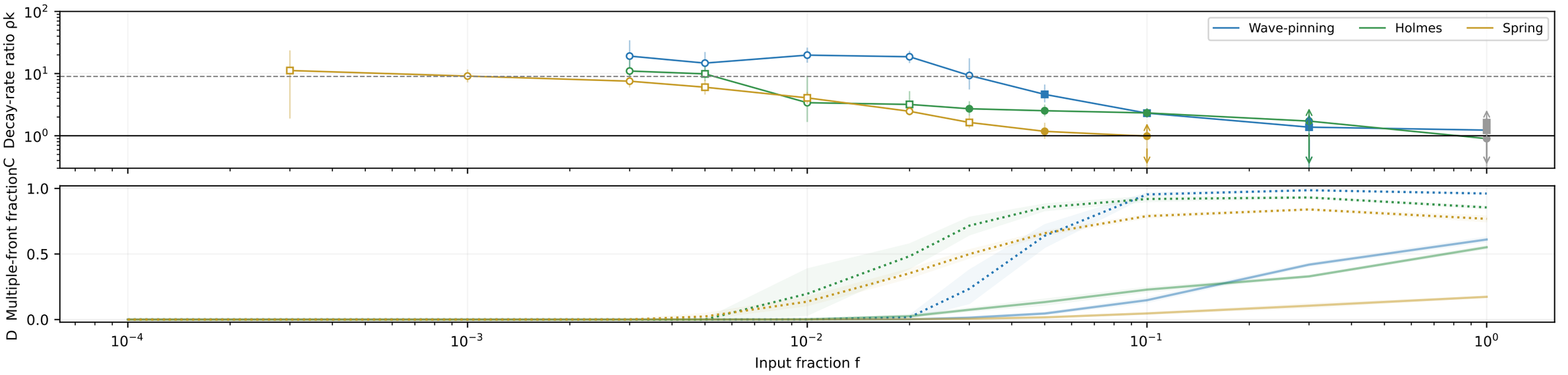
Open triangle + arrow: lower bound. ? Both times censored.
Top ticks: each size's antipodal-placement gap. ×: unavailable handover.

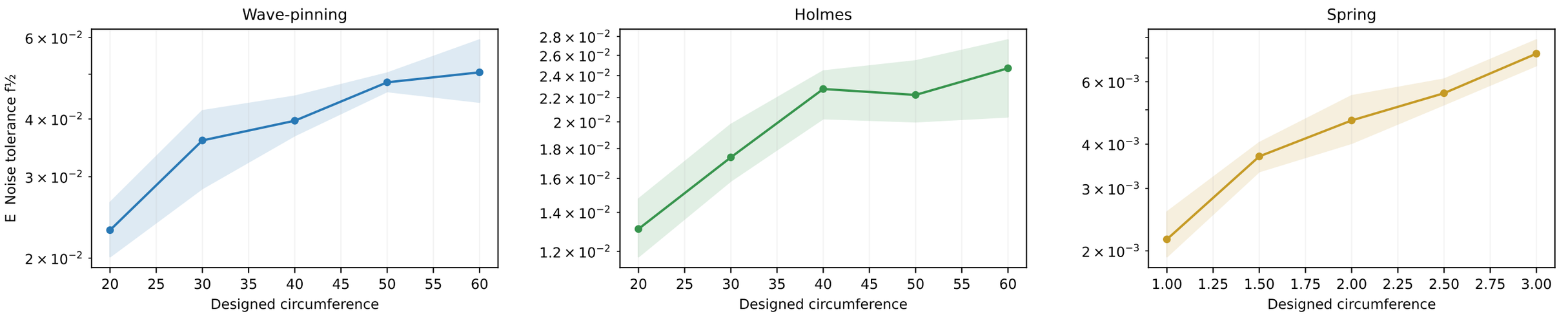
Enlarged cells hold heading longer under fluctuating input
At or above geometry at weak input; the advantage declines as input strengthens

A Native activity histories









Small: lighter solid; large: darker dotted. C: open fill single front; filled multiple fronts; grey fill low contrast; square >1.5× lag drift.
Pointwise whole-seed 95% CIs. Horizontal references: ratio 1 and nominal geometry 9. D is descriptive; it does not establish a cause of erosion.
H1 uses a post hoc f < large-cell f½ domain; frozen H1 remains inconclusive. Goryachev retained in supplement only.